7. VM Networking Modes (VirtualBox)

The networking mode decides who the virtual machine can talk to: the internet, the host, other VMs, or the whole home network.

Mode	VM to internet	VM to other VMs	VM to host	Outside to VM	Best used for
NAT	Yes	No	No	No	Quick internet access for one VM, updates and downloads. Each VM is isolated.
NAT Network	Yes	Yes	No (by default)	No	A lab where Kali, Ubuntu and Windows need to talk to each other and also get updates
Host-only	No	Yes	Yes	No	A fully isolated attack and defence lab with no internet
Bridged	Yes	Yes	Yes	Yes	When the VM must act like a real device on your network, like a server others need to reach

Short explanation of each

- **NAT:** the VM hides behind the host's IP, like a phone using a hotspot. It can go out but nothing can come in.
- **NAT Network:** like NAT, but multiple VMs share the same private network and can see each other.
- **Host-only:** a private network only between the host and the VMs. No internet at all.
- **Bridged:** the VM gets its own IP from your actual router, so it appears as a separate device on your home or college Wi-Fi.

Why vulnerable labs must never be bridged

- A vulnerable VM (Metasploitable, DVWA, old Windows) has known weaknesses, often with default passwords and open services.
- In bridged mode it sits directly on your real network, so anyone on that Wi-Fi can reach it and attack it.
- If it gets compromised, the attacker can move from that VM to your real laptops, phones and router.
- Attack tools running from Kali in bridged mode could also accidentally scan other people's devices on a shared network, which can be illegal.
- Safe approach: use **Host-only** or **NAT Network**, and add internet only when needed.